Performance and Cost Optimization in Smart LLM Routing

An Empirical Analysis of Anchor Routing Algorithms, Cascade Overhead, and Escalation Signal Specificity

Prepared by: Gemini Notebook | **Scope:** Synthesis of 5 Benchmark Studies on Model Routing & Cascades

Executive Summary

Key Research Takeaways

Model routing directs user prompts to the most cost-effective AI model without sacrificing accuracy. Across five benchmark studies evaluating routing algorithms on RouterBench and ArvanCloud model pools, the **Anchor method** emerges as the most practical solution—executing 24x faster than GPU-based methods (0.64 ms on CPU) while matching top accuracy. Furthermore, complex setups like **cascading models** (escalating unagreed responses) are frequently redundant, increasing cost and latency without net accuracy gains over a strong standalone router. The primary leverage point for system optimization is refining **escalation specificity**—correctly identifying when a cheap model's response is acceptable rather than triggering false escalations.

1. Benchmarking Nine Routing Methods (RouterBench)

Study 1 evaluated nine routing strategies across 11 models and 36,481 prompts (7,296 held out) using real dollar costs on RouterBench. All methods were cost-matched against the floor model (GPT-4) so differences reflect prediction quality.

Method	Accuracy Score	Cost / Query	Cost Savings	vs. Floor (GPT-4)	Decision Latency
Oracle (Cheapest Correct)	91.43%	$0.000230	93.0%	+13.32 pp	N/A
Router-BERT	78.74%	$0.002476	24.0%	+0.63 pp [+0.24, +1.03]	15.2 ms (GPU)
Token-Budget	78.70%	$0.002708	17.0%	+0.59 pp [+0.24, +0.94]	1.55 ms
Anchor	**78.67%**	**$0.002751**	**16.0%**	**+0.57 pp [+0.22, +0.91]**	**0.64 ms (CPU)**
Meta-Routing	78.61%	$0.002782	15.0%	+0.50 pp [+0.19, +0.81]	1.60 ms
KNN	77.81%	$0.002483	24.0%	–0.29 pp [–0.75, +0.16]	17.0 ms
Floor (GPT-4 Single)	78.11%	$0.003270	0.0% (Baseline)	0.00 pp	Instant
Random Baseline	51.75%	$0.000849	74.0%	–26.36 pp	Instant

Table 1: Routing benchmarks on RouterBench. Anchor ties Router-BERT quality while deciding 24x faster.

2. Real List Pricing vs. Token Proxy (ArvanCloud Pool)

Study 2 evaluated 7 models on 150 MMLU-Pro questions (90 held out) using OpenRouter list prices (2026-09-22). Token counts were a misleading cost proxy: prompt length + 4x completion weight spans 4.9x, whereas real list prices span **86x** and invert cost ordering.

Routing Option / Model	Accuracy (± SD)	Cost per Query ($)
Always GPT-5.6-Sol	93.1 ± 1.3%	$0.001755
Anchor Top-3 + Cascade	92.9 ± 1.9%	$0.001338
Matrix-Factorization	92.0 ± 2.2%	$0.001974
Anchor Alone	90.4 ± 2.5%	$0.000866
Always DeepSeek-V4-Flash	89.5 ± 2.1%	$0.000106
Random Choice	88.7 ± 2.1%	$0.001432

Table 2: Accuracy and cost on ArvanCloud pool over 12 splits. Frontier reduces to two models (GPT-5.6-Sol & DeepSeek-V4-Flash).

3. Training Data Density Explains Router Performance

> **Data Density vs. Task Diversity:** Study 3 showed Anchor beat weighted coin flips on RouterBench (+5.16 pp over convex hull, winning at 98% of operating points) but lagged on ArvanCloud (–0.62 pp, winning at 7%). Anchor does not learn topic suitability; it learns *which prompts are easy enough that a cheap model succeeds*. RouterBench provided 29,185 training prompts (128 anchors), while ArvanCloud provided only 60 training prompts (8 anchors / 7 per cluster), failing to establish sufficient density.

4. Cascade Mechanisms vs. Standalone Routers

Study 4 evaluated cascading policies (shortlisting top 3 anchors, calling two cheapest, and escalating on disagreement). Across 23 head-to-head comparisons at matched spend, **standalone Anchor outperformed Anchor + Cascade in every single test** (+5.29 pp vs –1.59 pp vs coin flips). Cascades substitute for a poor router rather than enhancing a good one; combining an accurate router with cascading adds pure cost overhead (2.5x calls).

5. Deconstructing the Escalation Signal (Specificity vs. Sensitivity)

Study 5 isolated the escalation decision by holding the shortlist fixed and varying the wrongness signal.

Escalation Signal Mechanism	Accuracy vs. Anchor	Cost vs. Anchor	Calls / Query
Agreement, then Weighted Vote	–0.73 pp	2.22x	2.50
Agreement, then Anchor's Pick	–1.58 pp	1.66x	2.26
Perfect Wrongness Signal (Ideal)	+10.84 pp	0.87x	1.34

Table 3: Escalation signal mechanics. Perfect signal gains +10.84 pp while cutting costs by 13%.

- **Sensitivity** P(escalate | wrong) buys *accuracy* (0.5 → 1.0 is worth +10.4 pp).
- **Specificity** P(stay | right) buys *cost savings* (1.0 → 0.5 increases bill from 0.88x to 1.60x).
- **Agreement Flaw:** Agreement sensitivity is 0.743 (at target), but specificity is only **0.619**, escalating 38% of correct answers for no benefit.
- **Target:** Hold sensitivity near 0.75 and push **specificity from 0.62 to 0.75**.

6. Methodological Caveats & Scope Limitations

1. **Tariffs:** OpenRouter list prices (2026-09-22) served as stand-ins for ArvanCloud.
2. **Sample Variance:** 90 held-out questions entail ~±6 pp variance on single figures.
3. **Logprobs:** Confidence-based cascading (e.g. BiLD) requires per-token logprobs, available on only 1 of 9 evaluated models.

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